Architecture Vision — Target State

Document Purpose

This document defines the **target-state architecture** for the platform that will replace the legacy CTM/Twilio call tracking system and evolve into a scalable, AI-powered legal services platform.

Scope: Future state design, technology decisions, and implementation roadmap. For the legacy system baseline, see ARCHITECTURE.md. For inefficiency analysis and redesign rationale, see ARCHITECTURE_OPPORTUNITIES.md.

Executive Summary

The strategic goal is to help DienerLaw become the largest immigration law firm in the United States, and to offer cost, speed, and quality levels that no other firm can match. They also help DienerLaw achieve best-in-class profitability and margins because of size, scale, automation, and AI.

The architecture described in this document improves operational effectiveness, increases efficiencies, reduces technical limitations, and enables a strategic shift towards **MyLegalTeam** — a subscription-based legal services platform targeting 1 million customers at $100/year, creating a **$100 million annual recurring revenue stream** on top of the firm's existing legal services income.

MyLegalTeam

MyLegalTeam gives every subscriber a persistent, personalized legal workspace — accessible via web and mobile apps — that fundamentally changes how immigration clients interact with their attorneys and legal advisors:

- **Document scanning.** Clients can scan or take call-phone photos of key documents which are then pixel-corrected and translated into markdown so the AI can answer questions and engest the information into the system.
- **Case organization.** Clients manage all immigration matters in one place: documents, forms, deadlines, and filing status — organized automatically, not buried in email threads and paper folders.

- **AI legal assistant.** A specialized immigration AI engine that knows each client's specific situation — their case history, filing status, family composition, and jurisdiction — and answers legal questions in context, not generically.
- **Attorney coordination.** Every interaction with attorneys and paralegals is logged, searchable, and tied to the relevant case. Clients see what was discussed, what was filed, and what comes next — eliminating the "what's happening with my case?" call.
- **Law and policy alerts.** When immigration laws change, court rulings shift precedent, or USCIS updates processing times, subscribers receive personalized alerts explaining how the change affects *their* specific situation — not a generic newsletter.

This platform is not a replacement for legal counsel — it is the delivery mechanism that makes DienerLaw's legal services dramatically more accessible, organized, and valuable than any competitor can offer.

Why this matters competitively. No immigration law firm offers anything like this today. The web and mobile apps create differentiation that cannot be replicated by hiring more attorneys or buying more ads. They build customer stickiness: clients who organize their legal life inside MyLegalTeam do not switch firms. And they drive repeat business — when a subscriber's spouse, parent, or sibling needs immigration help, the referral is automatic because the family already lives inside the platform.

Revenue model. The $100/year subscription revenue is *additive* to existing per-case legal fees. A client paying $100/year for the platform is also paying thousands for legal representation. The subscription covers the AI assistant, case organization tools, and alert service; legal work remains billable. At scale, the subscription revenue alone exceeds total current firm revenue — and the platform's stickiness increases the lifetime value of every legal engagement.

How we get there. The three horizons described in this document build toward MyLegalTeam incrementally:

- **Horizon 1** replaces the legacy vendor stack (CTM, Twilio, Zoho, Flow Legal, Snowflake) with owned infrastructure — RustPBX, Telnyx, and PostgreSQL — saving ~$450K/yr and establishing the technical foundation.
- **Horizon 2** adds AI-powered operations at the firm level: real-time transcription, agent coaching, automated intake, and intelligent routing — proving the AI capabilities that will later face customers.
- **Horizon 3** packages those capabilities into the MyLegalTeam consumer platform — the subscription app that scales to millions.

Everything in this architecture — every SIP trunk, every database schema, every AI pipeline — is designed to serve Horizon 3. The infrastructure decisions made in Horizon 1 are not just cost savings;

they are the foundation of a $100M ARR platform.

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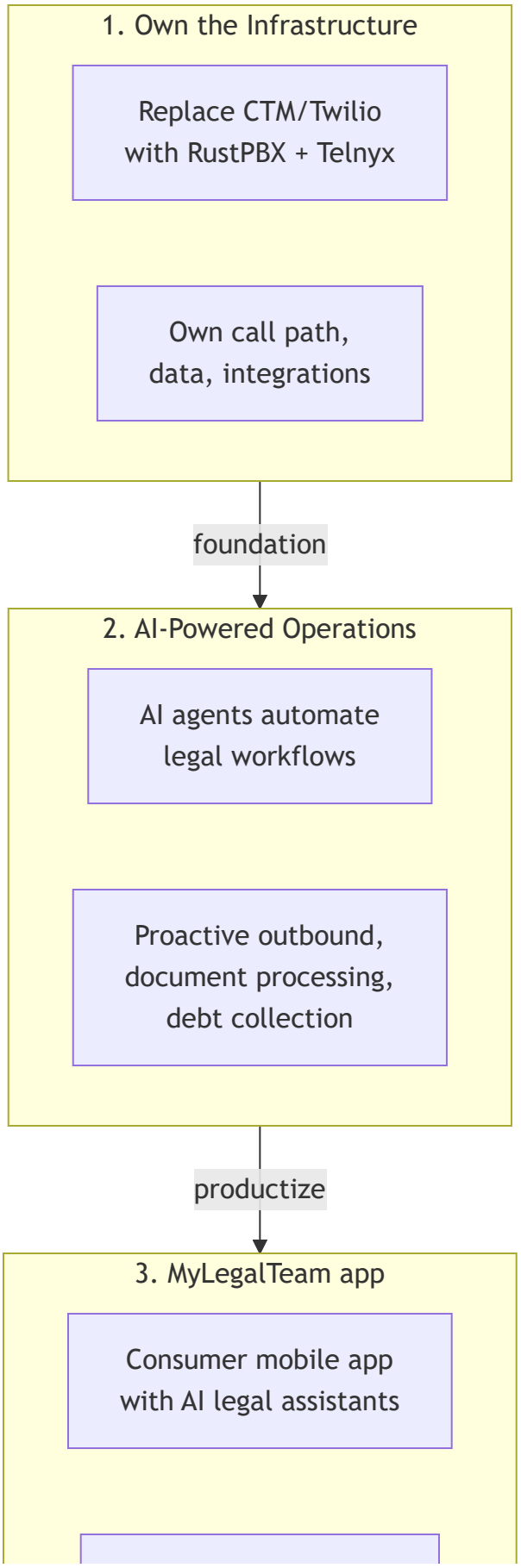
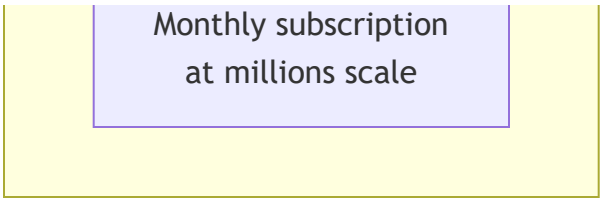
1. North Star: MyLegalTeam

The infrastructure work described in this document is Phase 1 of a three-horizon evolution that transforms the firm from a transactional legal practice into **MyLegalTeam** (working name) — a subscription-based, AI-powered legal services platform capable of serving millions of customers.


1. Own the Infrastructure
Replace CTM/Twilio with RustPBX + Telnyx
Own call path, data, integrations
foundation
2. AI-Powered Operations
AI agents automate legal workflows
Proactive outbound, document processing, debt collection
productize
3. MyLegalTeam app
Consumer mobile app with AI legal assistants



Horizon 1: Own the Infrastructure

Replace CTM and Twilio with a self-hosted RustPBX platform connected directly to Telnyx SIP trunking. Own the call path, call data, recordings, transcriptions, and all downstream integrations to Zoho CRM and Flow Legal Management. Eliminate vendor lock-in and double billing. **This is the focus of this document.**

Horizon 2: AI-Powered Operations (Future)

Deploy AI agents on the backend to organize incoming information, convert legacy documents into legal data and forms, orchestrate call center staff proactive workloads (outbound campaigns, customer information collection, debt collection), and oversee legal workflows for attorneys and paralegal assistants. Automate as much operational overhead as possible to increase firm capacity without proportional headcount growth.

Vendor replacement: Horizon 2 also replaces the remaining SaaS vendors — Zoho CRM, Flow Legal Management, and Snowflake — with custom Rust-based programs backed by PostgreSQL. The Zoho and Flow Legal integrations built in Horizon 1 (§11) are transitional: they keep the firm operational during the migration, but the long-term architecture consolidates all CRM, case management, and analytics into a single self-hosted Rust backend with PostgreSQL as the unified data store.

Horizon 3: MyLegalTeam Consumer Platform (Future)

Launch a customer-facing mobile app and subscription service providing: a client portal with all legal information and attorney communications; AI agents that help customers input information, answer attorney questions, and cross-reference legal documents; proactive updates when laws change; and ongoing automated legal guidance. Revenue model shifts from per-case transactions to monthly subscriptions at millions-of-users scale.

Business Model Evolution

```
TODAY (Legacy)                        HORIZON 3 (MyLegalTeam)
──────────────────────────────────    ──────────────────────────────────
Transactional legal factory           Ongoing automated legal service
Per-case revenue                      Monthly subscription revenue
Manual intake + processing            AI-orchestrated workflows
Attorney-dependent scaling            AI-agent scaling
Hundreds of active cases              Millions of subscribers
Call center as cost center            Call center as onboarding engine
CTM/Twilio vendor lock-in             Fully self-hosted platform
Zoho CRM + Flow Legal + Snowflake     Custom Rust + PostgreSQL backend
Data fragmented across vendors        Unified data under full ownership
```

> **Horizon 2 and 3 will be detailed in future revisions of this document.** The remainder focuses on Horizon 1.

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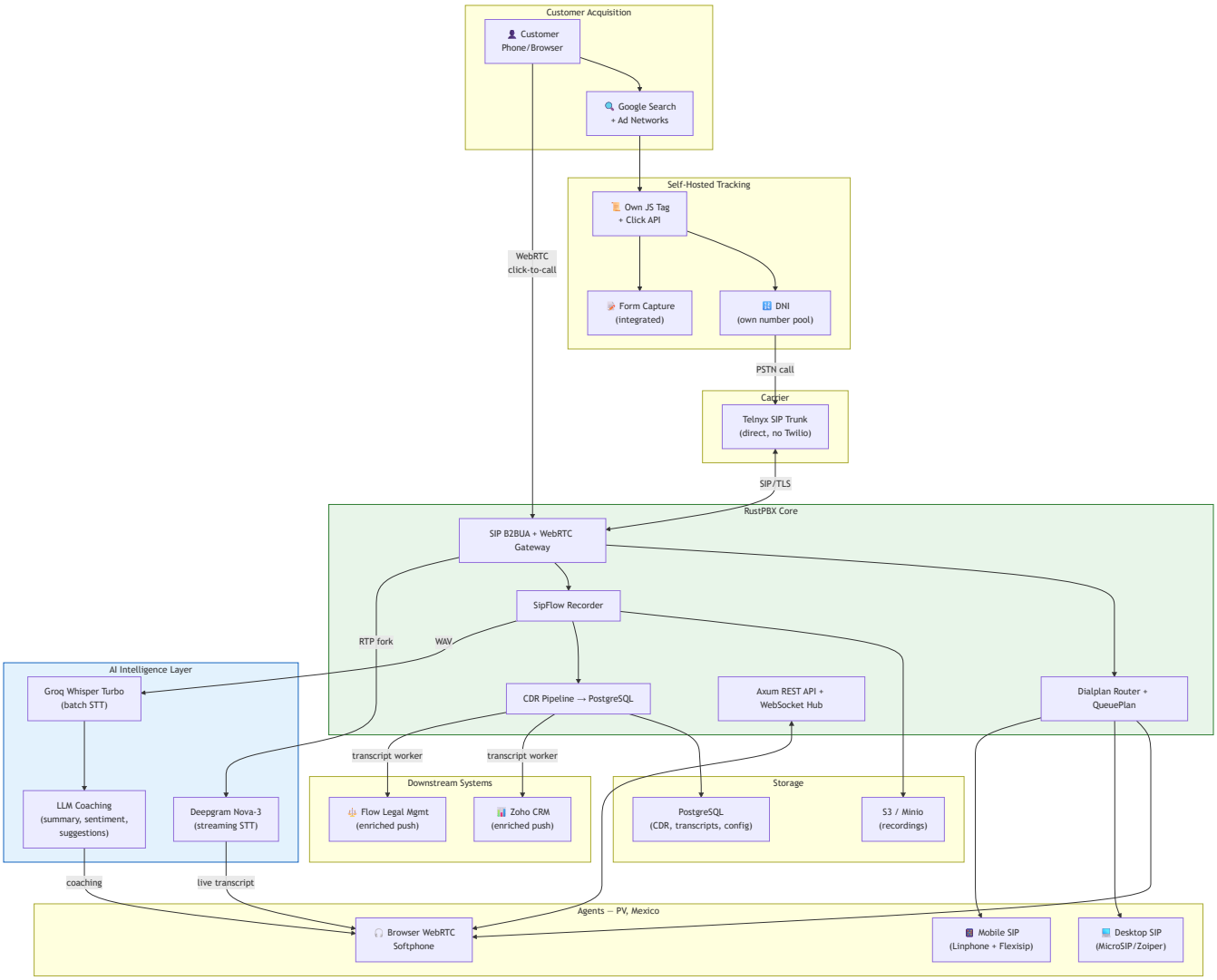
2. Horizon 1: System Overview

RustPBX replaces CTM as the central platform. It connects directly to Telnyx (eliminating Twilio), routes calls to agents, records and transcribes them, provides real-time AI coaching, and pushes enriched data to Zoho CRM and Flow Legal Management.



What Changes from Legacy

Aspect	Legacy (ARCHITECTURE.md)	Target (this document)
PSTN carrier	Twilio (via CTM)	Telnyx (direct)
Call platform	CTM (vendor SaaS)	RustPBX (self-hosted)
Click tracking	CTM JavaScript + API	Self-hosted JS tag + API
DNI numbers	CTM-managed (Twilio pool)	Self-managed (Telnyx pool)
Recording	CTM cloud (locked)	SipFlow → S3/Minio (owned)
Transcription	CTM native + Groq supplemental	Groq batch + Deepgram streaming
Agent coaching	None	LLM-powered (post-call + real-time)
CRM push	CTM → Zoho (basic)	RustPBX → Zoho (enriched)
Case mgmt push	CTM → Flow Legal (basic)	RustPBX → Flow Legal (enriched)
Agent UI	CTM browser widget	Leptos WASM console
Cost (50K min/mo)	~$2,000–5,000+/mo	~$725/mo

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3. Telnyx Integration: Pure SIP Trunking

Decision: Telnyx operates exclusively as a PSTN carrier. No webhooks, no Call Control API, no cloud recording, no Telnyx-managed transcription. RustPBX owns all call intelligence.

SIP Connection Configuration

Parameter	Value
Connection Type	Credential-based (dynamic IP, supports NAT)
Auth Method	SIP REGISTER with username/password
Outbound Proxy	sip.telnyx.com:5060 (UDP/TCP) or :5061 (TLS)
Transport	TLS preferred; UDP acceptable behind NAT with STUN
Codecs (priority)	1. PCMU (G.711μ) 2. PCMA (G.711A) 3. G.722 4. Opus
DTMF	RFC 2833 (in-band RTP events)
RTP Port Range	16384–32768 (configure firewall accordingly)
Outbound Voice Profile	Created in Telnyx Mission Control; applied to connection
Webhook URL	**NONE** — leave blank to stay in SIP trunking mode

> ⚠️ **Critical**: Setting a webhook URL on a Telnyx SIP Connection converts it to programmable voice mode and can anchor media in distant regions. Leave it blank.

RustPBX Trunk Config (rustpbx.toml)

```
[trunks.telnyx]
server = "sip.telnyx.com"
transport = "tls"
port = 5061
username = "<telnyx_sip_username>"
password = "<telnyx_sip_password>"
register = true
register_expires = 300
codecs = ["PCMU", "PCMA", "G722"]
outbound_proxy = "sip.telnyx.com"
```

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4. Softphone & Agent Endpoint Architecture

Decision: Hybrid Model — Agents Choose Their Audio Path

The Leptos WASM web console is **always** the command-and-control interface for every agent. It displays queue state, live transcripts, coaching notes, CDR history, and provides call control buttons. The audio path is flexible — agents select the endpoint that best fits their environment.

> **Agent Location:** The call center operates from **Puerto Vallarta, Mexico**. All SIP signaling and RTP media traverse the US-Mexico internet path, adding approximately **30–80ms round-trip latency**. Codec selection prioritizes resilience to jitter (Opus with FEC enabled). Agents rely on consumer-grade Mexican ISPs — internet failover strategy is a critical infrastructure concern.

Puerto Vallarta Network Architecture

Latency profile:

Path Segment	Typical RTT	Notes
Agent → Mexican ISP PoP	5–15ms	Last-mile, consumer broadband
Mexican ISP → US border crossing	15–30ms	Guadalajara/Mexico City IX → McAllen/Laredo
US border → Telnyx SIP edge	10–25ms	US backbone
Total SIP signaling RTT	**30–70ms**	Within ITU-T G.114 target (< 150ms one-way)
Total RTP media one-way	**15–35ms**	Well within conversational quality thresholds

Codec selection (optimized for Mexico path):

Priority	Codec	Bandwidth	Why
1	Opus (WebRTC)	24–48 kbps	Built-in FEC, adaptive bitrate, jitter resilience
2	G.722 (desktop SIP)	64 kbps	Wideband quality, good jitter tolerance
3	G.711μ / PCMU	64 kbps	Fallback — no FEC, sensitive to packet loss

Internet failover strategy:

Layer	Primary	Failover	Detection	RTO
Internet link	Wired broadband (Telmex/Izzi)	Telcel 4G/5G hotspot	Router WAN failover	30–60s
SIP registration	Auto-re-register on IP change	Multi-device ring	REGISTER expiry (300s)	5–10s
Active call	WebRTC ICE restart (browser)	Call routes to other device	ICE connectivity check	2–5s
Agent status	Auto-away on heartbeat timeout	Supervisor dashboard alert	WebSocket heartbeat (15s)	15–30s

> **Recommendation:** Equip each agent workstation with USB tethering-capable phone (Telcel plan) and configure router for automatic WAN failover. Prefer browser-mode WebRTC softphone (Opus FEC) over desktop SIP for jitter resilience.

Three Endpoint Modes

Mode	Transport	Audio Path	Control Path	Platforms
Browser (default)	SIP-over-WSS (WebRTC)	Browser microphone → RustPBX WebRTC GW	Leptos console (same tab)	All
Desktop SIP	SIP/TLS + SRTP	MicroSIP / Zoiper → RustPBX registrar	Leptos console (separate browser tab)	Win / Mac
Mobile SIP	SIP/TLS + SRTP via Flexisip	Linphone → Flexisip push GW → RustPBX	Leptos console (optional)	iOS / Android

Recommended Software per Mode

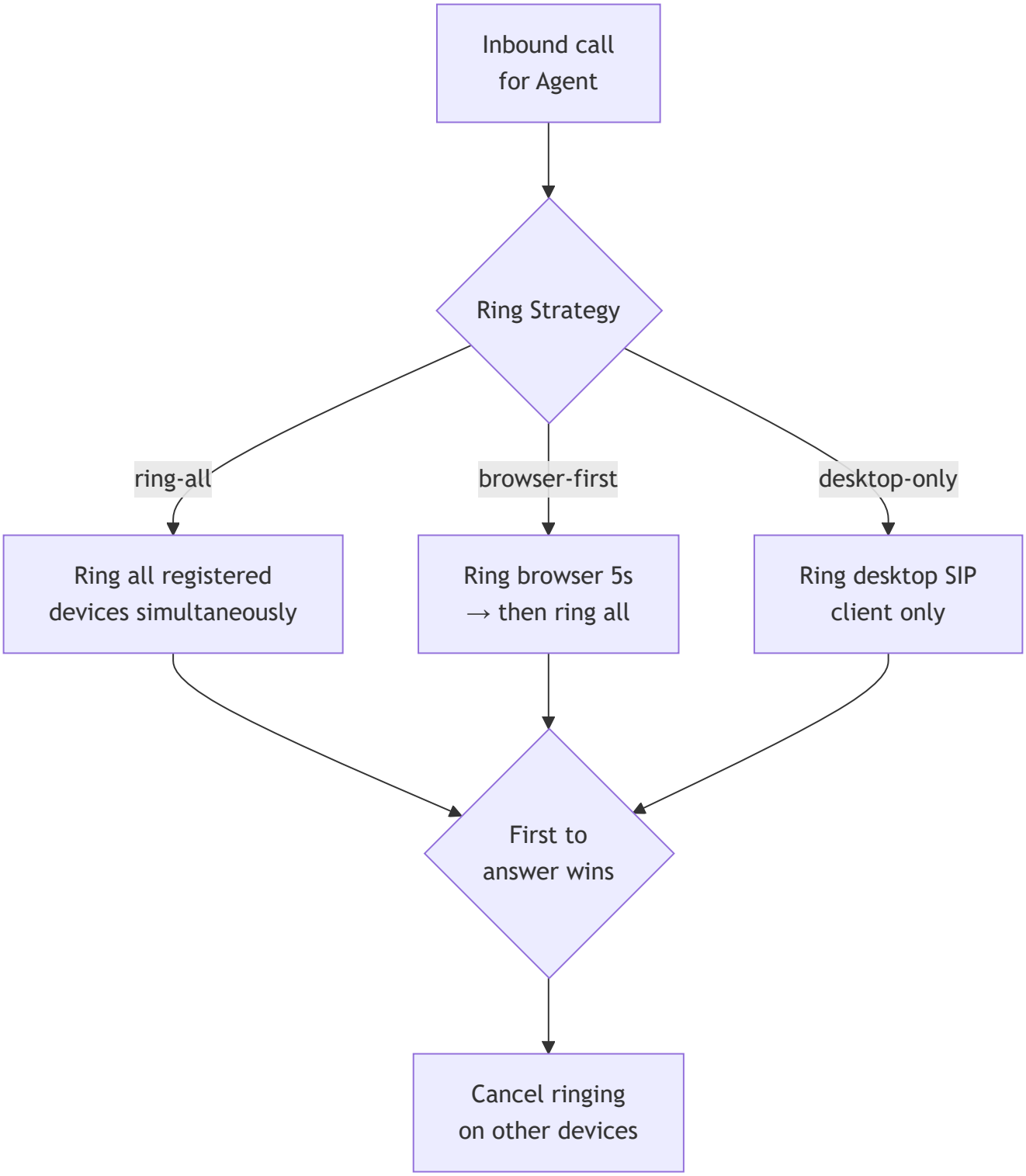
Platform	Software	License	Notes
Windows desktop	MicroSIP	Open source (free)	SIP/TLS/SRTP, compact
Mac desktop	Zoiper 5 PRO	~€40/seat one-time	SIP/TLS/SRTP, polished UI
iOS	Linphone	GPLv3 (free)	CallKit integration, push wake
Android	Linphone	GPLv3 (free)	Firebase push wake
All platforms	Browser (Leptos WebRTC)	Free	Default, zero install

Softphone Cost (50 agents)

Item	Cost
MicroSIP (Windows)	$0 (open source)
Zoiper 5 PRO (Mac)	~€40 × Mac count (one-time)
Linphone (iOS/Android)	$0 (open source)
Browser WebRTC	$0 (built-in)
Flexisip push gateway	$0 (self-hosted, AGPLv3)

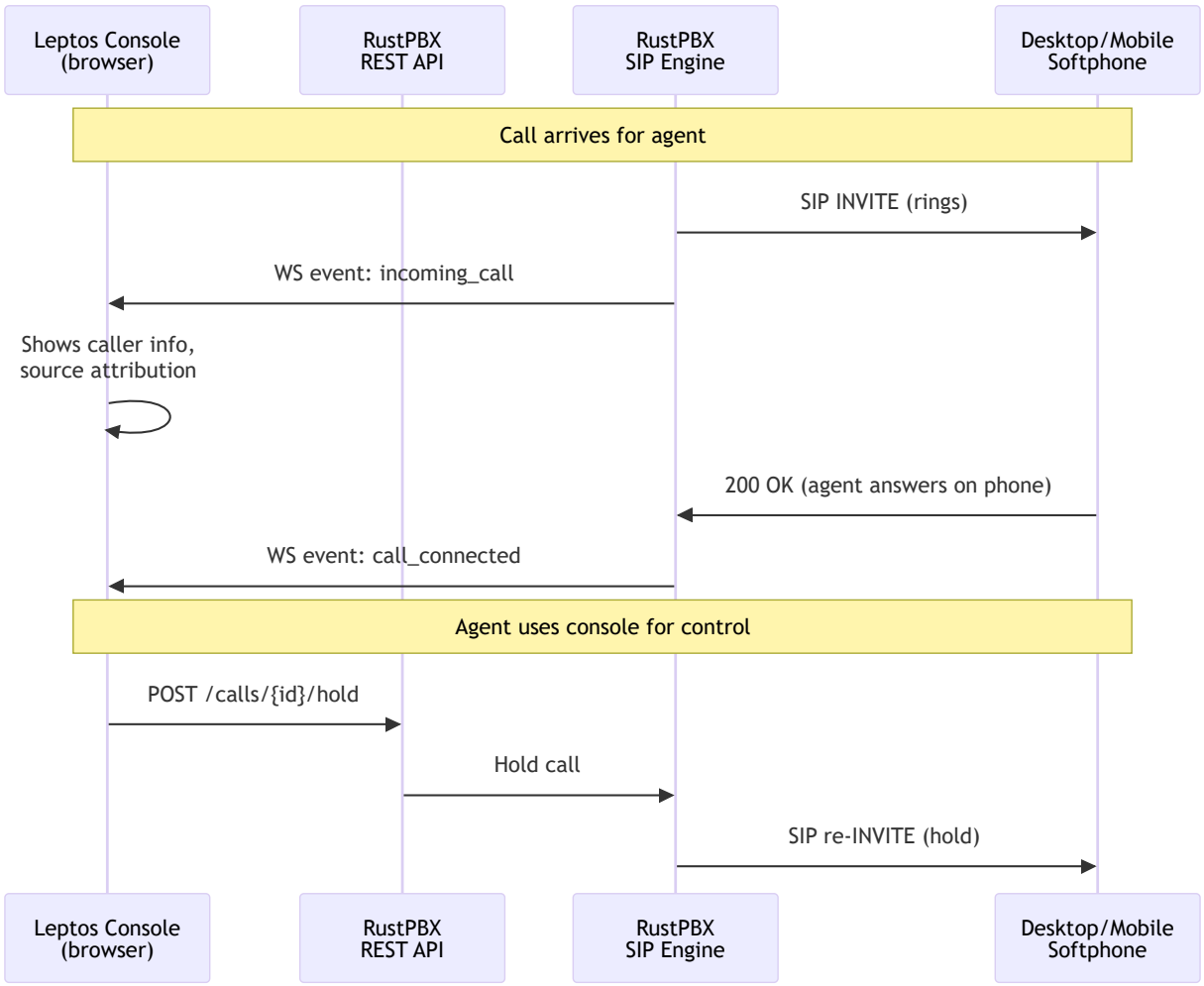
Multi-Device Ring Strategy

When a call arrives for an agent, RustPBX can ring multiple registered devices simultaneously or in sequence, configurable per agent/queue:



Desktop/Mobile Control Flow

When an agent uses a desktop SIP client or mobile phone for audio, the Leptos browser console remains the control surface. Call control commands flow through the REST API:



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5. Database: PostgreSQL

Decision: PostgreSQL via sqlx

PostgreSQL is the right fit for a VoIP platform workload: concurrent read/write (agents generate CDRs while supervisors run reports), JSONB columns for variable metadata, full-text search for transcripts via `tsvector` / `tsquery`, LISTEN/NOTIFY for event-driven transcript pipeline, and `sqlx` is Tokio-native with compile-time query verification.

sqlx Integration

```rust
// rustpbx-shared/src/db.rs
use sqlx::postgres::PgPool;

pub async fn connect(database_url: &str) -> Result<PgPool, sqlx::Error> {
    PgPool::connect(database_url).await
}

// Compile-time verified query:
let cdr = sqlx::query_as!(
    CdrRecord,
    r#"SELECT id, call_id, caller, callee, start_time, duration,
              recording_path, transcript_id, coaching_json
       FROM cdrs WHERE call_id = $1"#,
    call_id
).fetch_one(&pool).await?;
```

Key Tables

```
-- CDR with JSONB for flexible metadata
CREATE TABLE cdrs (
    id              BIGSERIAL PRIMARY KEY,
    call_id         UUID NOT NULL UNIQUE,
    caller          TEXT NOT NULL,
    callee          TEXT NOT NULL,
    direction       TEXT NOT NULL,        -- 'inbound' | 'outbound' | 'internal'
    start_time      TIMESTAMPTZ NOT NULL,
    answer_time     TIMESTAMPTZ,
    end_time        TIMESTAMPTZ,
    duration_secs   INTEGER,
    disposition     TEXT,                 -- 'answered' | 'no-answer' | 'busy' | 'fa
    agent_id        TEXT,
    queue_name      TEXT,
    recording_path  TEXT,
    transcript_id   BIGINT REFERENCES transcripts(id),
    metadata        JSONB DEFAULT '{}',   -- source DID, CRM data, tags
    coaching        JSONB,                -- LLM analysis results
    created_at      TIMESTAMPTZ DEFAULT NOW()
);
CREATE INDEX idx_cdrs_start ON cdrs(start_time DESC);
CREATE INDEX idx_cdrs_agent ON cdrs(agent_id, start_time DESC);
CREATE INDEX idx_cdrs_caller ON cdrs(caller);

-- Transcripts with full-text search
CREATE TABLE transcripts (
    id              BIGSERIAL PRIMARY KEY,
    call_id         UUID NOT NULL REFERENCES cdrs(call_id),
    provider        TEXT NOT NULL,        -- 'groq' | 'deepgram'
    content         TEXT NOT NULL,
    word_timestamps JSONB,
    diarization     JSONB,
    search_vector   TSVECTOR GENERATED ALWAYS AS (to_tsvector('english', content)) S
    created_at      TIMESTAMPTZ DEFAULT NOW()
);
CREATE INDEX idx_transcripts_search ON transcripts USING GIN(search_vector);

-- Voicemail
```

```
CREATE TABLE voicemails (
    id              BIGSERIAL PRIMARY KEY,
    extension_id    TEXT NOT NULL,
    caller          TEXT NOT NULL,
    recording_path  TEXT NOT NULL,
    transcript      TEXT,
    is_read         BOOLEAN DEFAULT FALSE,
    created_at      TIMESTAMPTZ DEFAULT NOW()
);

-- Click tracking sessions
CREATE TABLE click_sessions (
    id              BIGSERIAL PRIMARY KEY,
    session_id      UUID NOT NULL UNIQUE,
    source          TEXT,                 -- 'google' | 'facebook' | 'instagram'
    campaign        TEXT,
    keyword         TEXT,
    gclid           TEXT,
    fbclid          TEXT,
    landing_url     TEXT,
    tracking_number TEXT,
    visitor_ip      INET,
    created_at      TIMESTAMPTZ DEFAULT NOW()
);

-- Integration outbox (transactional outbox pattern)
CREATE TABLE integration_outbox (
    id              BIGSERIAL PRIMARY KEY,
    cdr_id          BIGINT REFERENCES cdrs(id) NOT NULL,
    target_system   TEXT NOT NULL,        -- 'zoho_crm' | 'flow_legal'
    payload         JSONB NOT NULL,
    status          TEXT DEFAULT 'pending',
    attempts        INT DEFAULT 0,
    last_attempt_at TIMESTAMPTZ,
    last_error      TEXT,
    created_at      TIMESTAMPTZ DEFAULT NOW(),
    sent_at         TIMESTAMPTZ
);
```

```
CREATE INDEX idx_outbox_pending ON integration_outbox (target_system, status)
    WHERE status IN ('pending', 'failed');
```

Deployment

Environment	Database	Notes
Development	Docker: `postgres:16-alpine`	Local, disposable
Staging	Supabase free tier or Docker	Mirrors production schema
Production	Managed PostgreSQL (RDS/Supabase/DO) or self-hosted	Backups, replication

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6. Transcription Architecture: Dual-Provider Strategy

Decision: Groq Whisper Turbo (batch) + Deepgram Nova (streaming)

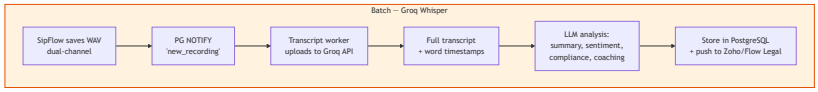
Each engine serves a distinct use case. Groq provides the cheapest high-accuracy batch transcription for post-call records. Deepgram provides the only viable real-time streaming transcription for live agent coaching.

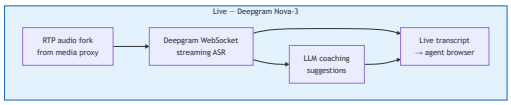
Head-to-Head Comparison

Dimension	Groq Whisper Large v3 Turbo	Deepgram Nova-2	Deepgram Nova-3
Pricing	**$0.04/hour** ($0.00067/min)	$0.0043/min	$0.0077/min streaming
Cost at 50K min/mo	**$33**	$215	$385
Speed	228× real-time (batch)	29.8s per hour (batch)	Sub-300ms streaming
Real-time streaming	❌ No	✅ WebSocket	✅ WebSocket
Accuracy (WER)	~10–12%	~8.4%	~6.84%
Speaker diarization	❌	✅ Native	✅ Native
Custom vocabulary	Prompt-only (224 tokens)	✅ Custom model	✅ Keyterm Prompting
API style	OpenAI-compatible (file upload)	REST + WebSocket	REST + WebSocket

> **Never use Telnyx STT passthrough** — it resells Deepgram at 3.5× markup ($0.015/min vs. $0.0043/min direct).

Transcription Pipeline





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7. Web UI: Leptos + DaisyUI + WASM Agent Console

Existing Framework

RustPBX ships with a built-in Axum web server. The current upstream console uses Alpine.js + MiniJinja templates at `/console/`. We replace it with a Leptos 0.8 WASM SPA using the `leptos-daisyui-rs` component library (80+ DaisyUI components already implemented in Leptos).

Component	Technology	Notes
UI Framework	Leptos 0.8	Rust → WASM, reactive, fine-grained updates
Component Library	leptos-daisyui-rs	80+ pre-built DaisyUI components
CSS	DaisyUI (Tailwind-based)	Theme switching, responsive
Build Tool	Trunk	WASM compilation + asset bundling
WebSocket	Built-in RustPBX WS hub	Events, call state, live transcript
API	RustPBX REST API (Axum)	Call control, CDR, config

Agent Console Layout

```
┌──────────────────────────────────────────────────────────────────────┐
│  [Logo]  Agent: Maria G.  ● Online  ▼Queue: Intake    🔔 3 VMs      │
├──────────────┬──────────────────────────────┬──────────────────────┤
│              │                              │                      │
│  QUEUE       │     ACTIVE CALL              │  SIDEBAR             │
│  PANEL       │                              │                      │
│              │  ☎ +1-555-123-4567           │  📝 Live Transcript  │
│  ● Agent 1   │  Source: Google Ads - PI_Natl │  ──────────────────  │
│    Idle      │  Duration: 4:32              │  Caller: "I was in   │
│  ● Agent 2   │                              │  an accident last ... "│
│    On Call   │  [Answer][Hold][Mute]        │                      │
│  ○ Agent 3   │  [Transfer][Notes][End]      │  🤖 AI Suggestions   │
│    Offline   │                              │  ──────────────────  │
│              │                              │  "Ask about medical  │
│  Waiting: 2  │                              │   treatment timeline"│
│  Avg: 0:45   │                              │                      │
│              │                              │  📊 Sentiment: 😐    │
├──────────────┴──────────────────────────────┴──────────────────────┤
│  [CDR History]  [Voicemail (3)]  [Settings]  [Supervisor Dashboard]│
└──────────────────────────────────────────────────────────────────────┘
```

Key Components to Build

Component	leptos-daisyui-rs Base	Custom Logic
WebRTC softphone	—	Full custom: SIP-over-WS, getUserMedia, ICE
Call control dock	`button`, `button_group`	REST API integration
Queue dashboard	`table`, `badge`, `indicator`	WebSocket live updates
Live transcript	`chat`	Deepgram WS → text stream rendering
Coaching sidebar	`card`, `alert`	LLM suggestion display
CDR viewer	`table`, `pagination`, `input`	PostgreSQL full-text search
Voicemail inbox	`table`, `badge`	Audio player + transcript
Settings	`form`, `select`, `toggle`	Agent preferences

Shared Rust Types (WASM ↔ Server)

```rust
// rustpbx-shared/src/models.rs — compiled to both server and WASM
#[derive(Serialize, Deserialize, Clone)]
pub struct CallEvent {
    pub call_id: Uuid,
    pub event_type: CallEventType,
    pub caller: String,
    pub callee: String,
    pub timestamp: DateTime<Utc>,
    pub metadata: Option<serde_json::Value>,
}

#[derive(Serialize, Deserialize, Clone)]
pub enum CallEventType {
    Ringing, Answered, Held, Resumed, Transferred, Ended,
    TranscriptChunk(String),
    CoachingSuggestion(String),
}
```

---

8. Call Recording & Storage

Decision: RustPBX SipFlow handles all recording. No Telnyx cloud recording.

Pipeline

```
Inbound/Outbound Call
    │
    ▼
SipFlow Recorder (unified SIP + RTP capture)
    │
    ├──→ WAV file (dual-channel: caller L, agent R)
    │     stored: /recordings/YYYY/MM/DD/{call_id}.wav
    │
    ├──→ CDR record → PostgreSQL
    │     includes: recording_path, duration, channels
    │
    └──→ Post-call event triggers transcript pipeline:
          1. Upload WAV → Groq Whisper Turbo API
          2. Transcript JSON → PostgreSQL
          3. LLM analysis → summary, sentiment, coaching → JSONB
          4. Push enriched data → Zoho CRM + Flow Legal
```

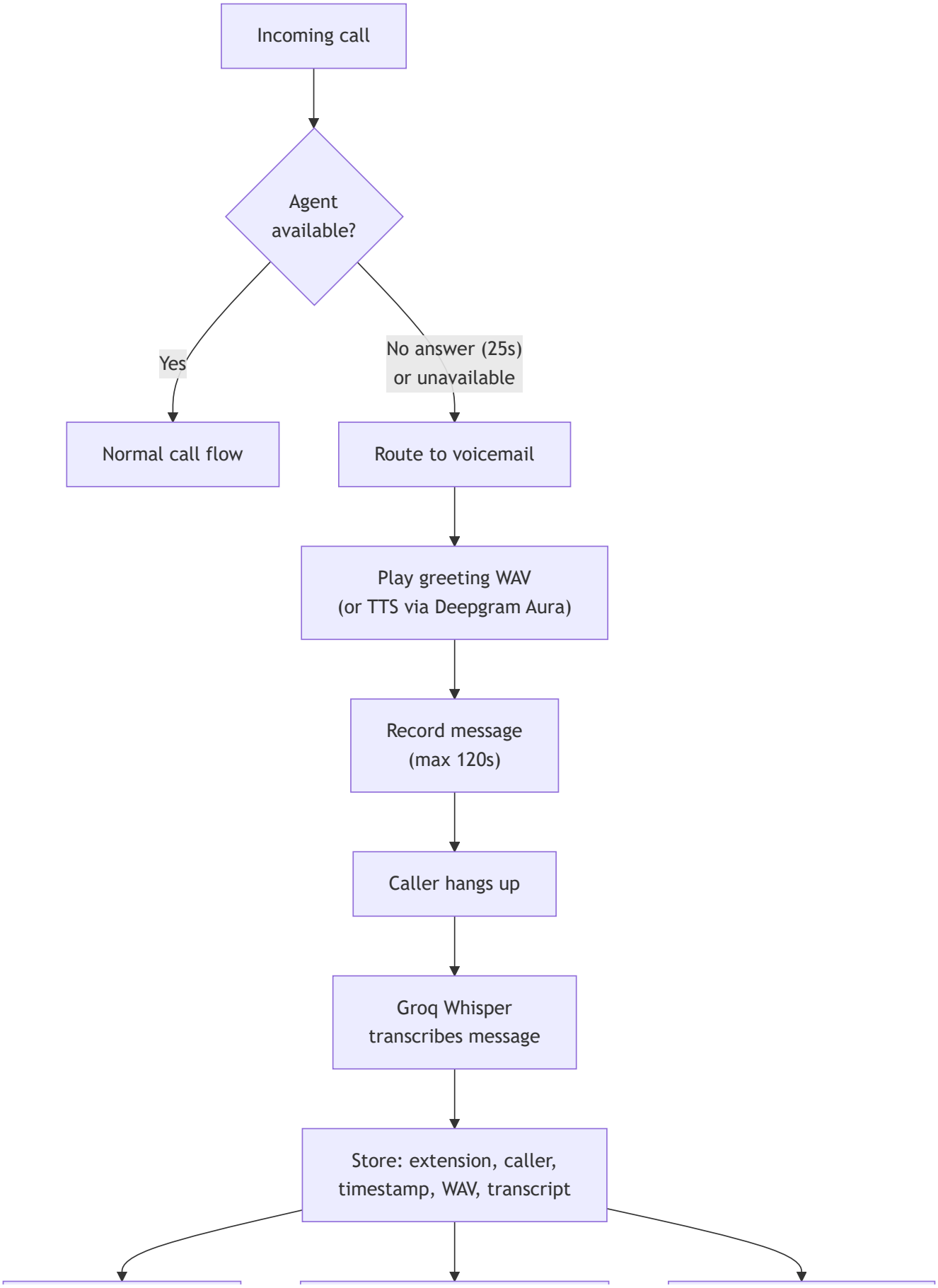
Storage Backend Options

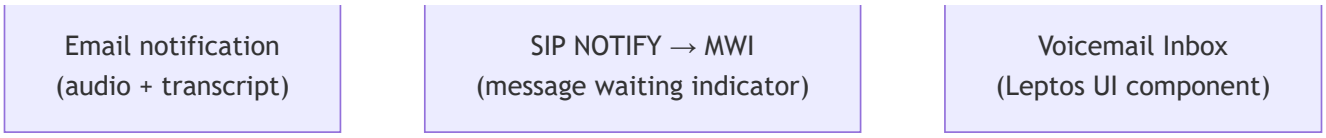
Backend	Config Key	Use Case
Local disk	`recording.backend = "local"`	Development, single-server
S3 / Minio	`recording.backend = "s3"`	Production, multi-server
SCP / SFTP	`recording.backend = "sftp"`	Compliance archive

9. Voicemail System (Custom Build)

Neither RustPBX nor Telnyx provides turnkey voicemail. Built as a dialplan extension + addon.

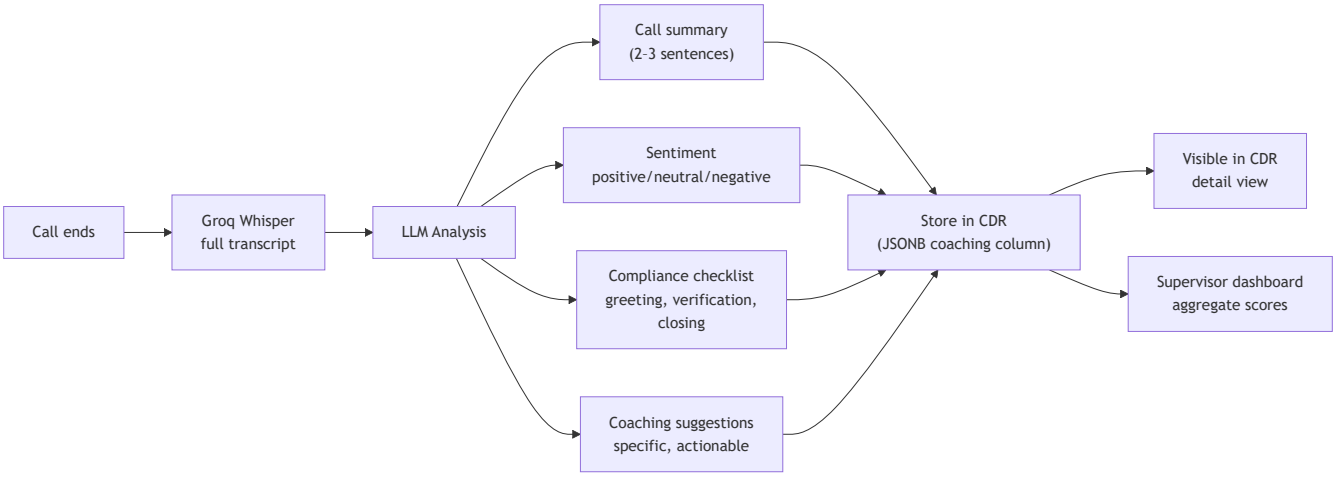
Voicemail Flow


Incoming call
Agent available?
Yes
No answer (25s) or unavailable
Normal call flow
Route to voicemail
Play greeting WAV (or TTS via Deepgram Aura)
Record message (max 120s)
Caller hangs up
Groq Whisper transcribes message
Store: extension, caller, timestamp, WAV, transcript

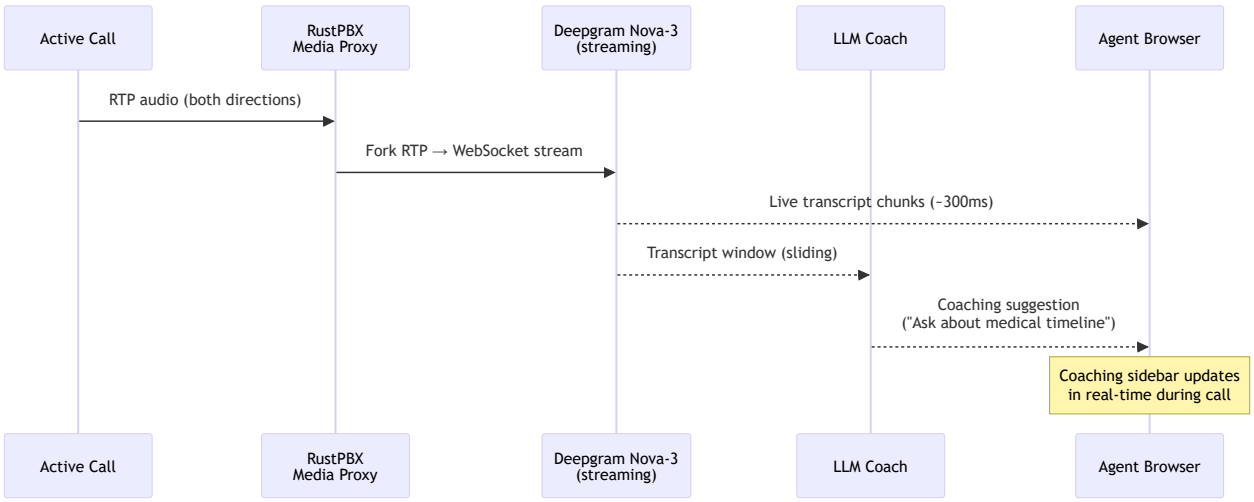


10. Agent Coaching Architecture

Phase 1: Post-Call Coaching (MVP)



Phase 2: Real-Time Coaching (Future)



> **Note:** Phase 2 requires intercepting RTP at the media proxy level and forking to an external WebSocket. Architecturally possible (RustPBX relays all RTP) but requires custom development.

> Telnyx Media Forking is NOT available in SIP trunk mode.

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11. External System Integrations

RustPBX replaces CTM as the integration hub. Both downstream pushes happen in the same transcript worker job, ensuring consistency.

11.1 Zoho CRM Integration

> **Transitional:** This integration keeps the firm operational during migration. In Horizon 2, Zoho CRM is replaced by a custom Rust CRM engine backed by PostgreSQL. The data model and enrichment logic built here carry forward.

The `rustpbx-transcript-worker` daemon pushes enriched call data to Zoho CRM via the Zoho REST API v2 after post-call processing.

Enriched payload (vs. legacy CTM push):

```
Call Ends → SipFlow saves WAV → PG NOTIFY 'new_recording'
    │
    ▼
rustpbx-transcript-worker:
    ├── 1. Groq Whisper Turbo → full transcript
    ├── 2. LLM analysis → summary, sentiment, disposition
    ├── 3. Correlate with click/form session (PostgreSQL JOIN)
    │
    ▼
POST https://www.zohoapis.com/crm/v2/Calls
    Body: {
        "Subject": "Inbound call from +1-555-123-4567",
        "Call_Duration": "4:32",
        "Call_Type": "Inbound",
        "Who_Id": "<contact_id>",
        "Description": "<AI-generated summary>",
        "Transcript_URL": "https://pbx.example.com/api/v1/transcripts/12345",
        "Recording_URL": "https://pbx.example.com/api/v1/recordings/12345.wav",
        "Sentiment": "Positive",
        "Ad_Source": "Google Ads - Campaign: PI_National",
        "Landing_Page": "https://example.com/personal-injury",
        "Call_Disposition": "Qualified Lead"
    }
```

Parameter	Value
API Base	`https://www.zohoapis.com/crm/v2/`
Auth	OAuth 2.0 (refresh token grant, auto-renew every 55 min)
Module	`Calls` (standard) + custom fields
Rate Limit	100 req/min (Standard), 500/min (Enterprise)
Contact Matching	`GET /crm/v2/Contacts/search?phone={number}`

Custom fields to create in Zoho:

Field	Type	Purpose
`Transcript_URL`	URL	Link to searchable transcript
`Recording_URL`	URL	Link to recording playback
`AI_Summary`	Multi-line	LLM-generated call summary
`Sentiment_Score`	Picklist	Positive / Neutral / Negative / Escalation
`Ad_Source`	Single-line	Click attribution (campaign, ad group, keyword)
`Landing_Page`	URL	Page customer was on when they called
`Session_ID`	Single-line	Links call to click tracking session

11.2 Flow Legal Management Integration

> **Transitional:** This integration keeps the firm operational during migration. In Horizon 2, Flow Legal is replaced by a custom Rust case management engine backed by PostgreSQL. The enriched payload structure designed here informs the native data model.

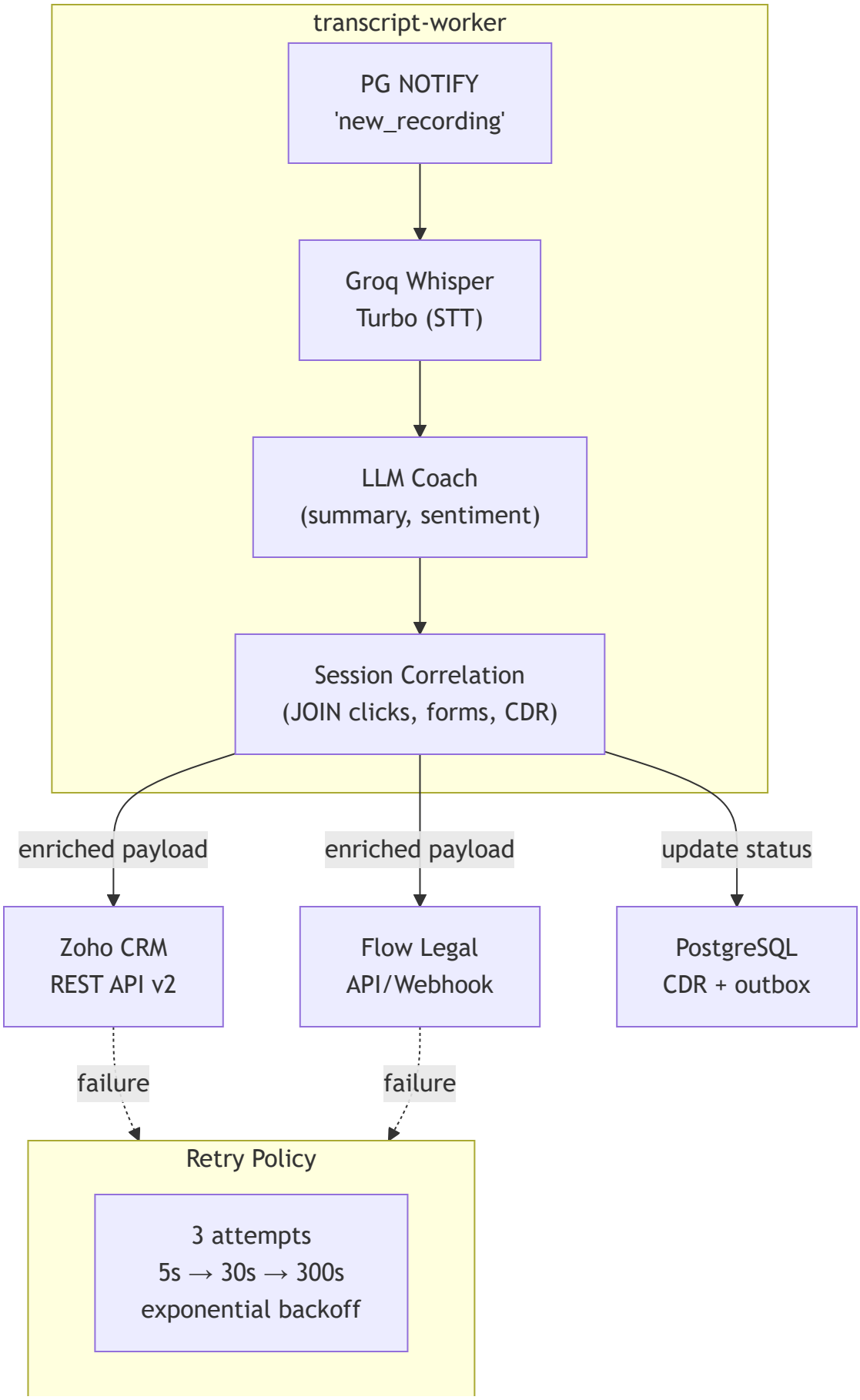
Same transcript worker job pushes enriched data to Flow Legal alongside Zoho.

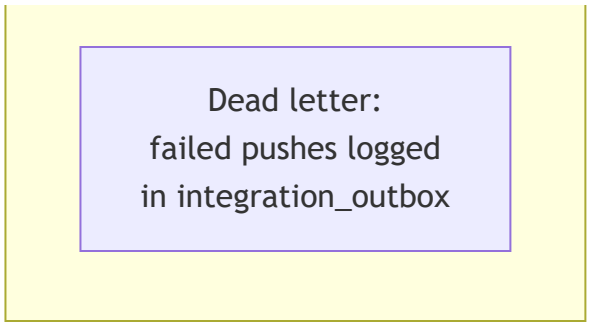
```
rustpbx-transcript-worker:
    ├── Push to Zoho CRM (§11.1)
    │
    └── Push to Flow Legal Management
            Body: {
                "caller_phone": "+1-555-123-4567",
                "caller_name": "Jane Doe",
                "call_datetime": "2026-02-21T14:30:00Z",
                "duration_seconds": 272,
                "recording_url": "https://pbx.example.com/…",
                "transcript_text": "<full transcript>",
                "ai_intake_summary": "Caller reports slip-and-fall at…",
                "case_type_suggestion": "Personal Injury - Premises Liability",
                "compliance_flags": ["HIPAA_mention_detected"],
                "disposition": "Qualified - Schedule Consultation",
                "ad_attribution": { "source": "Google Ads", "campaign": "PI_National
            }
```

Parameter	Value
API Base	TBD — requires Flow Legal API documentation review
Auth	API key or OAuth (TBD)
Case Matching	Match by phone number to existing case, or create intake

> ⚠️ **Action required:** Obtain Flow Legal Management API documentation to confirm endpoint structure, authentication method, and field mappings.

11.3 Unified Integration Data Flow


transcript-worker
PG NOTIFY
'new_recording'
Groq Whisper
Turbo (STT)
LLM Coach
(summary, sentiment)
Session Correlation
(JOIN clicks, forms, CDR)
enriched payload
enriched payload
update status
Zoho CRM
REST API v2
Flow Legal
API/Webhook
PostgreSQL
CDR + outbox
failure
failure
Retry Policy
3 attempts
5s → 30s → 300s
exponential backoff


Dead letter:
failed pushes logged
in integration_outbox

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12. Technology Stack Summary

Layer	Technology	Notes
Carrier	Telnyx SIP Trunking	Pure transport, credential auth, no Twilio
PBX Core	RustPBX (v0.3.15+)	Rust, SIP B2BUA, WebRTC gateway
Web Server	Axum (inside RustPBX)	REST API + WebSocket hub
Agent UI	Leptos 0.8 + DaisyUI + WASM	leptos-daisyui-rs (80+ components)
UI Build	Trunk	WASM compilation + asset bundling
WebRTC	Built-in RustPBX gateway	SIP-over-WS + ICE/DTLS/SRTP
Desktop Softphone (Win)	MicroSIP	Open source, SIP/TLS/SRTP
Desktop Softphone (Mac)	Zoiper 5 PRO	~€40/seat one-time
Mobile Softphone	Linphone	GPLv3, iOS + Android
Mobile Push Gateway	Flexisip	AGPLv3, Docker, APNs + FCM
Batch STT	Groq Whisper Large v3 Turbo	$0.04/hr, 228× realtime
Streaming STT	Deepgram Nova-2/3	$0.0043–0.0077/min, WebSocket
TTS	Deepgram Aura-2	$0.030/1K chars, voicemail greetings
LLM (coaching)	OpenAI / Groq LLM / local	Post-call + real-time analysis
Database	PostgreSQL 16+	Via sqlx (async, compile-time checked)
Recording Store	Local → S3/Minio (prod)	WAV files, lifecycle management
Shared Types	`rustpbx-shared` crate	Rust structs → server + WASM
CRM	Zoho CRM (REST API v2)	OAuth 2.0, enriched auto-push
Case Management	Flow Legal Management	API/webhook, case-call linking

13. Monthly Cost Projection

Assuming 50 agents, ~50,000 minutes/month:

Line Item	Provider	Unit Cost	Monthly Est.
SIP Trunking (inbound)	Telnyx	~$0.0085/min	$425
SIP Trunking (outbound)	Telnyx	~$0.005/min	$125
Phone numbers (50 DIDs)	Telnyx	~$1.00/mo each	$50
Post-call transcription	Groq Whisper Turbo	$0.04/hr	$33
Streaming STT (10% of calls)	Deepgram Nova-2	$0.0043/min	$22
LLM coaching analysis	Groq/OpenAI	~$0.001/call	$50
TTS (voicemail greetings)	Deepgram Aura-2	$0.030/1K chars	$5
Recording storage (S3)	AWS/Minio	~$0.023/GB	$15
PostgreSQL (managed)	Supabase/RDS	—	$0–25
Flexisip push gateway	Self-hosted	$0	$0
TOTAL			**~$725–750/mo**

One-time: ~€400 ($440) for Zoiper Mac licenses. All other softphones free.

Legacy comparison: CTM/Twilio typically $2,000–5,000+/month for similar volume.

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14. Implementation Phases (Horizon 1)

Phase 1: Foundation (Weeks 1–3)

- [ ] Configure Telnyx SIP Connection (credential auth, no webhook)
- [ ] Configure RustPBX trunk to Telnyx
- [ ] Basic dialplan: inbound DID → extension routing
- [ ] Verify SipFlow recording works end-to-end

- [ ] Set up PostgreSQL (Docker `postgres:16-alpine` for dev)
- [ ] Create sqlx migrations for cdrs, transcripts, voicemails, click_sessions, agents
- [ ] Create `rustpbx-shared` crate with core model types
- [ ] Scaffold `rustpbx-console` Leptos app with login + shell layout

Phase 2: Agent Console MVP (Weeks 4–6)

- [ ] WebRTC softphone component (SIP-over-WS to RustPBX) — browser mode
- [ ] Call control dock (answer, hold, mute, transfer, hangup)
- [ ] REST API call control for desktop/mobile agents
- [ ] WebSocket event channel for real-time state updates
- [ ] Queue dashboard with live agent status
- [ ] CDR viewer with search/filter/pagination (Postgres full-text search)
- [ ] Test MicroSIP (Win) and Zoiper (Mac) registration to RustPBX
- [ ] Multi-device ring strategy in dialplan

Phase 3: Transcription Pipeline (Weeks 7–8)

- [ ] Build `rustpbx-transcript-worker` daemon (PG LISTEN/NOTIFY)
- [ ] Groq Whisper Turbo integration for post-call batch STT
- [ ] Transcript storage in PostgreSQL with tsvector search index
- [ ] Transcript display in call detail view
- [ ] Voicemail system (record → transcribe → email notify → MWI)

Phase 4: Coaching & Analytics (Weeks 9–10)

- [ ] LLM post-call analysis (summary, sentiment, coaching → JSONB)
- [ ] Coaching notes display in CDR detail view
- [ ] Supervisor dashboard (aggregate metrics, agent performance)
- [ ] Recording playback with synchronized transcript

Phase 4b: External System Integrations (Weeks 10–11)

- [ ] Zoho CRM OAuth 2.0 token management (encrypted refresh token in PG)
- [ ] Create custom fields in Zoho CRM

- [ ] Zoho push in transcript worker (POST /crm/v2/Calls)
- [ ] Contact matching by phone number
- [ ] Obtain Flow Legal API documentation and confirm integration method
- [ ] Flow Legal push in transcript worker
- [ ] Integration outbox table + retry logic
- [ ] Supervisor dashboard: integration status panel
- [ ] Parallel CTM cutover testing (run RustPBX pushes alongside CTM)

Phase 5: Real-Time Features (Weeks 11–12)

- [ ] Deepgram Nova streaming integration (WebSocket)
- [ ] Live transcript panel during active calls
- [ ] Real-time coaching sidebar (LLM suggestions)
- [ ] Click tracking API + self-hosted DNI
- [ ] Form capture integration

Phase 6: Mobile & Production Hardening (Weeks 13–14)

- [ ] Deploy Flexisip push gateway (Docker, APNs + FCM)
- [ ] Configure Linphone for iOS (CallKit) + Android (Firebase)
- [ ] Test mobile push wake → register → ring flow end-to-end
- [ ] TLS everywhere (SIP-TLS, WSS, HTTPS)
- [ ] Authentication + authorization (agent/supervisor roles)
- [ ] S3/Minio for recording storage
- [ ] Monitoring, alerting, log aggregation
- [ ] Number porting from Twilio → Telnyx
- [ ] CTM decommissioning (after all integration validation passes)

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15. Key Architectural Decisions Log

#	Decision	Chosen	Rationale
1	Telnyx mode	Pure SIP Trunking	Full PBX control, no webhook latency, lowest cost
2	Call routing	RustPBX dialplan	Single routing authority, offline-capable
3	Recording	RustPBX SipFlow	Data sovereignty, no per-min cost
4	Batch STT	Groq Whisper Turbo	6.5× cheaper than Deepgram for post-call
5	Streaming STT	Deepgram Nova-2/3	Only option with WebSocket streaming + diarization
6	Agent UI	Leptos 0.8 + DaisyUI (WASM)	Existing library, shared Rust types
7	Softphone model	Hybrid — agent chooses	Browser WebRTC default; desktop/mobile SIP optional
8	Database	PostgreSQL via sqlx	JSONB, full-text search, LISTEN/NOTIFY, async
9	CRM integration	Zoho REST API v2	Replace CTM auto-push with enriched payload
10	Case mgmt integration	Flow Legal API	Replace CTM data feed with enriched payload
11	Click tracking	Self-hosted API	Eliminate CTM DNI dependency, cookie-independent
12	Long-term direction	MyLegalTeam platform	H1→H2→H3 evolution toward subscription model
13	Vendor independence	Replace all SaaS	Zoho, Flow Legal, Snowflake → Rust + PostgreSQL in H2

16. Cross-References

Document	Purpose
ARCHITECTURE.md	Legacy system baseline — how it works today
ARCHITECTURE_OPPORTUNITIES.md	Inefficiency analysis and five redesign opportunities
NORTH_STAR_CONCEPT.md	MyLegalTeam three-horizon vision (concept capture)
TESTING_PLAN_OF_ACTION.md	Nine-stage testing strategy
PROJECT_OVERVIEW.md	High-level project summary and RustPBX capabilities
ARCHITECTURE_VISION_DRAFT.md	Archived — raw material used to build this document